Copper-Nicus Mine Development

Team J8

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Problem Definition

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- There is growing demand for minerals causing the anticipated **critical mineral gap**
- British Columbia has significant copper deposits, providing an opportunity to create mines which can meet the demand

Introduction to Copper-Nicus Mine Development

Who are we, what is our goal?

We have been hired by **Copper-Nicus Mining Ltd.** to recommend a mine site for development.

We have selected the mineral deposit which will provide the best overall:

- Environmental protection
- Economic return
- Community benefit
- Respect for culturally significant land

We have chosen to recommend **Deposit 2 as an underground mine.**

Project Goals and Stakeholders


General Community
Government
Stakeholders
Indigenous Peoples
Mining Company

Common Stakeholder Needs

The needs identified are based on Environmental, Economic, and Social Sustainability

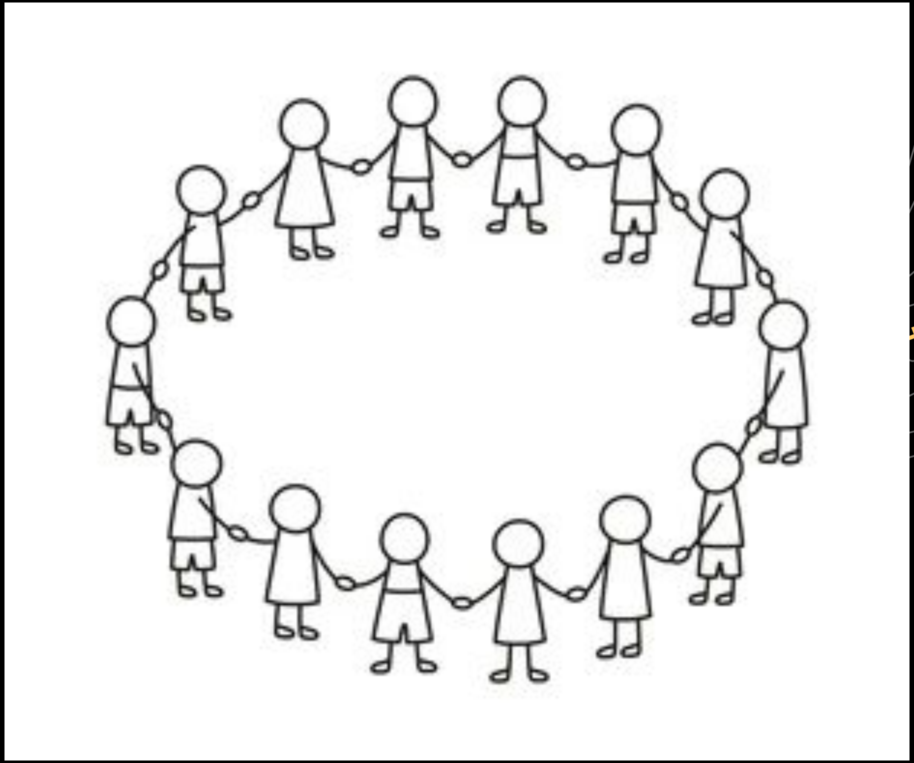
- Minimize Cost
- Create new Infrastructure
- Increase Job Opportunities
- Preserve Traditional Land

Stakeholder Engagement

To members of the Community:

Please feel free to…

- Join **public meetings**
- Look at our website for news
- Participate and join online discussion forums



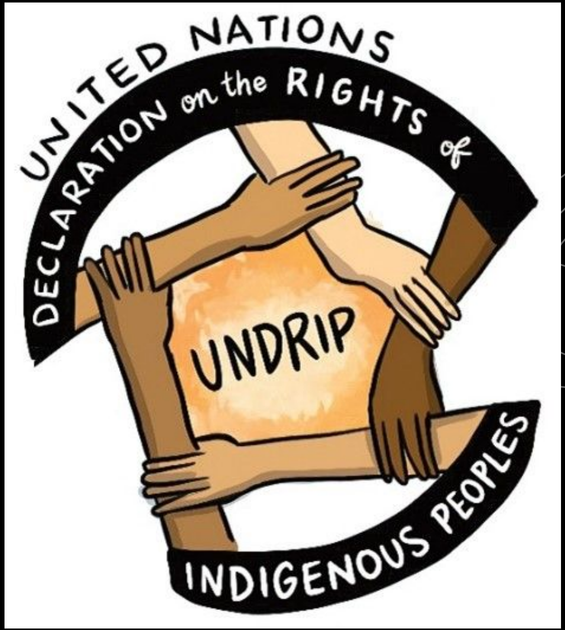
To Indigenous People:

Please…

- Attend **consultation** meetings
- Participate in community polls
- Join discussion forums

We want to learn about your

- Cultural and environmental priorities
- Knowledge of traditional land



Collaboration with the Government of BC

Government will:

- Submit formal project reports
- Participate in checkpoint meetings
- Discuss environmental assessments

They shall advise

- Safety and sustainability requirements
- Environmental regulations

Role of Mining Company

They will be collaborating throughout the project:

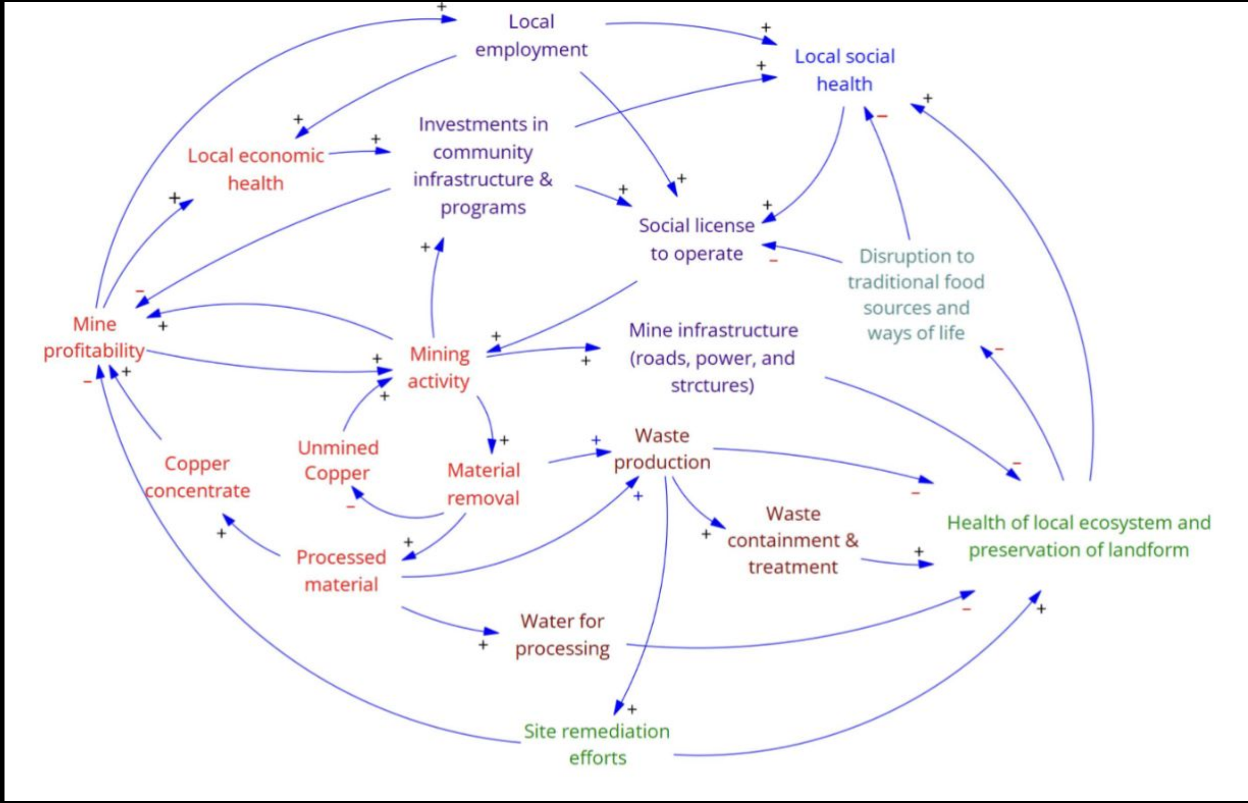
- Sharing formal reports and updates
- Holding planning and technical meetings
- Review project feasibility

Analysis of Mining Deposits

Analysis of Mining Deposits

- **Context**: Understanding The Complex System - Causal Loop Diagrams (CLD)
- Analysis from Map
- Block Models
- Streamlined Life Cycle Assessment (SLCAs)
- Final Scoring from Weighted Decision Matrix (WDMs)

Causal Loop Diagrams



Map of Mining Deposits

Deposit 1:

- Very Remote
- Major new infrastructure required

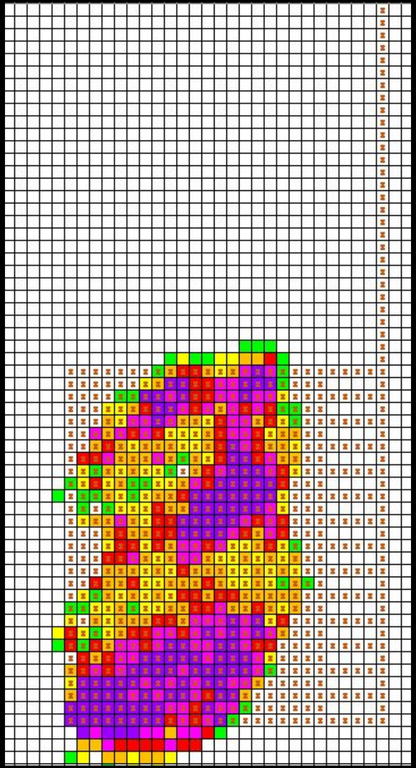
Deposit 2:

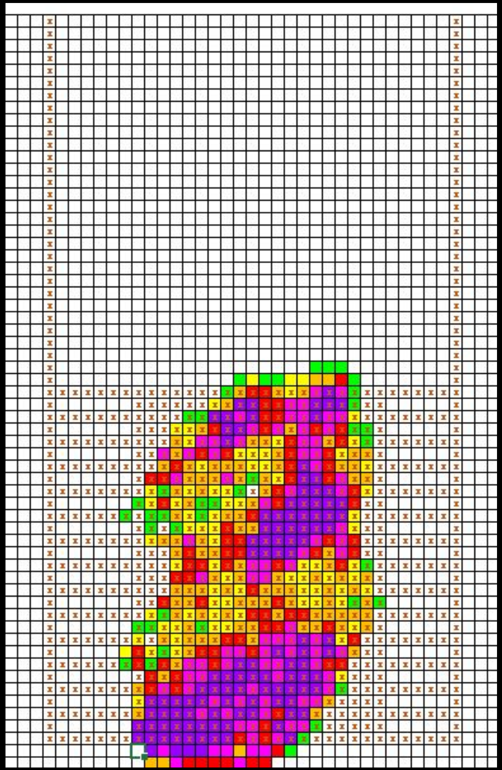
- Highway Accessible w/ Bridge
- Beside Nicus River (Runoff)

Deposit 3:

- Beside a town and archaeological site

Deposit 1 - Underground Mine



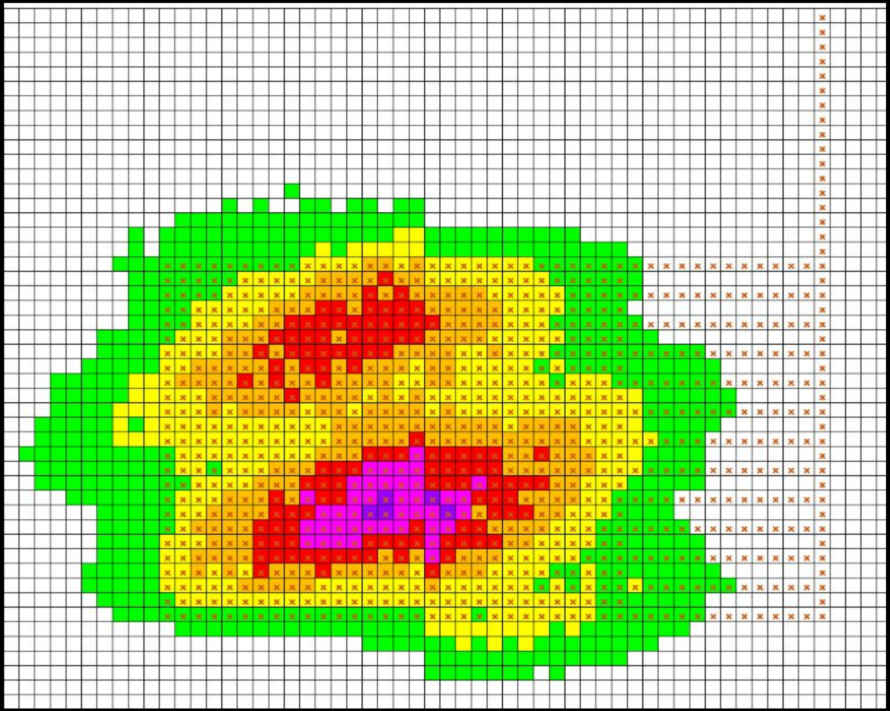


Overall Outputs (Single Shaft)

- Total Profit: **6.5 Billion**
- Waste Rock: 35 Million tonnes
- Copper Conc. (20%): 1.4 million tonnes

Mine Production Lifetime: 27 years

Deposit 2 - Underground Mine

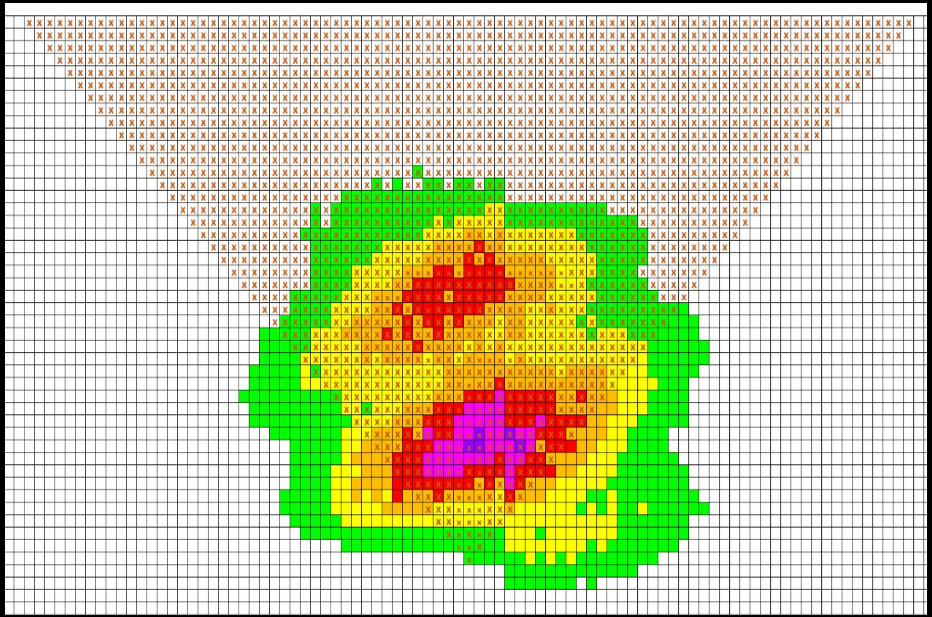


- Total Profit: **8.3 Billion**
- Waste Rock: 29 million tonnes
- Copper Conc. (20%): 2.1 million tonnes

Mine Production Lifetime:

57 years

Deposit 2 - Open Pit

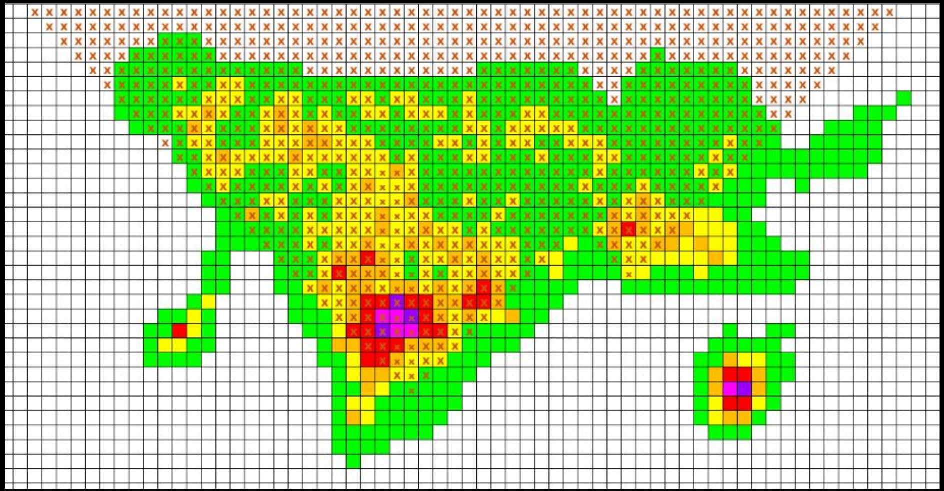


- Total Profit: **8.3 Billion**
- Waste rock: 246 million tonnes
- Copper Conc. (20%): 2.2 million tonnes

Comparison to Deposit 2 Underground

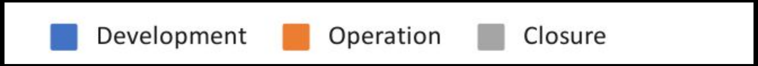
- **8.5x Waste rock for similar profit**

Deposit 3 - Open Pit

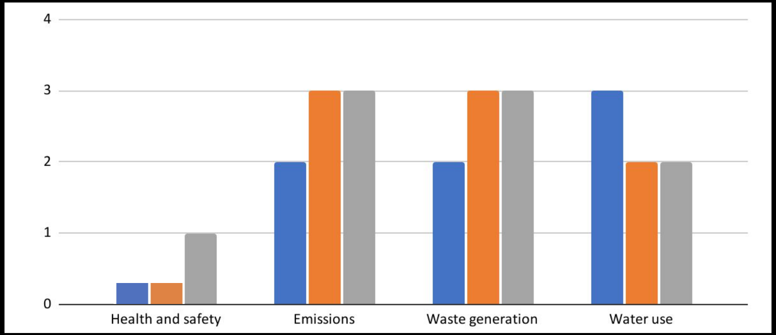


- Total Profit: **5.2 Billion**
- Waste rock: 70 million tonnes
- Copper Conc. (20%): 1.7 million tonnes

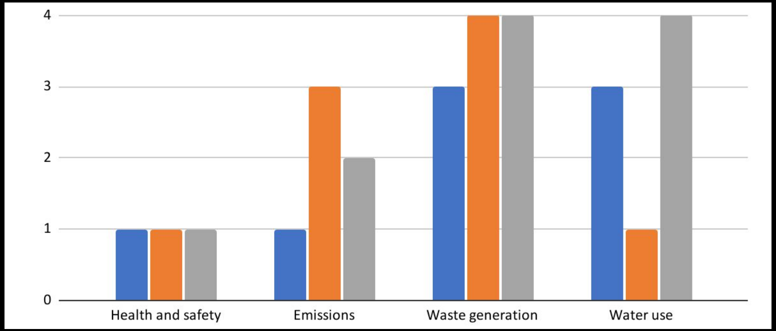
SLCA



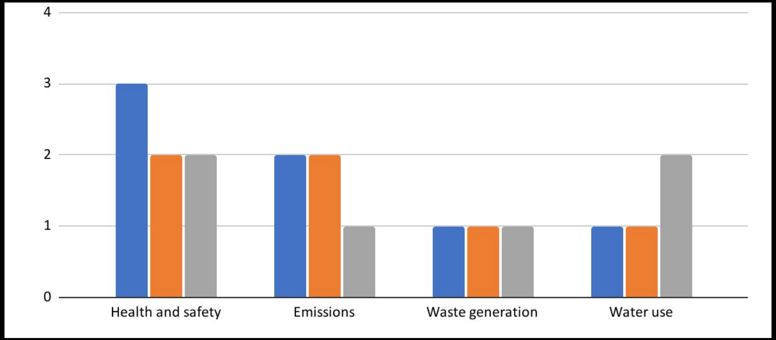
Deposit 1



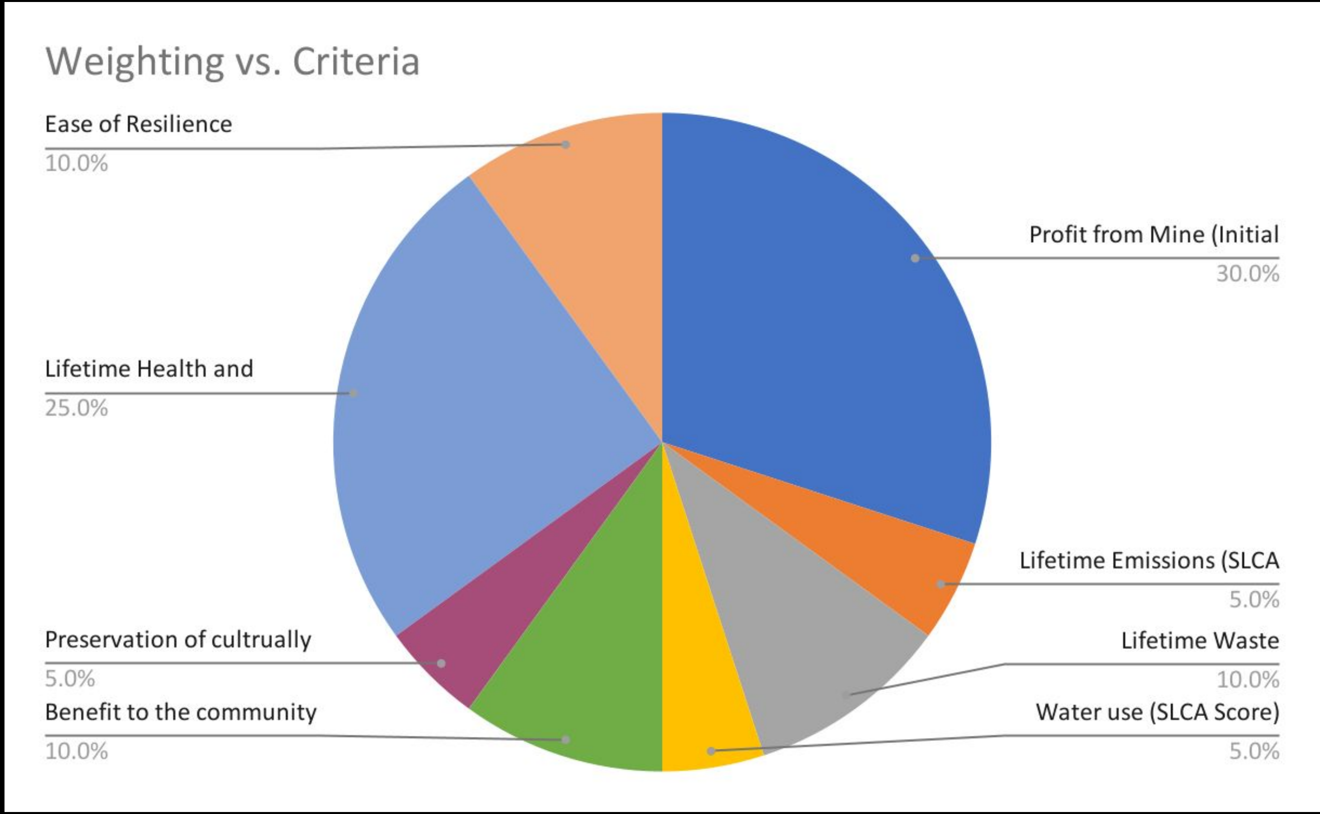
Deposit 2



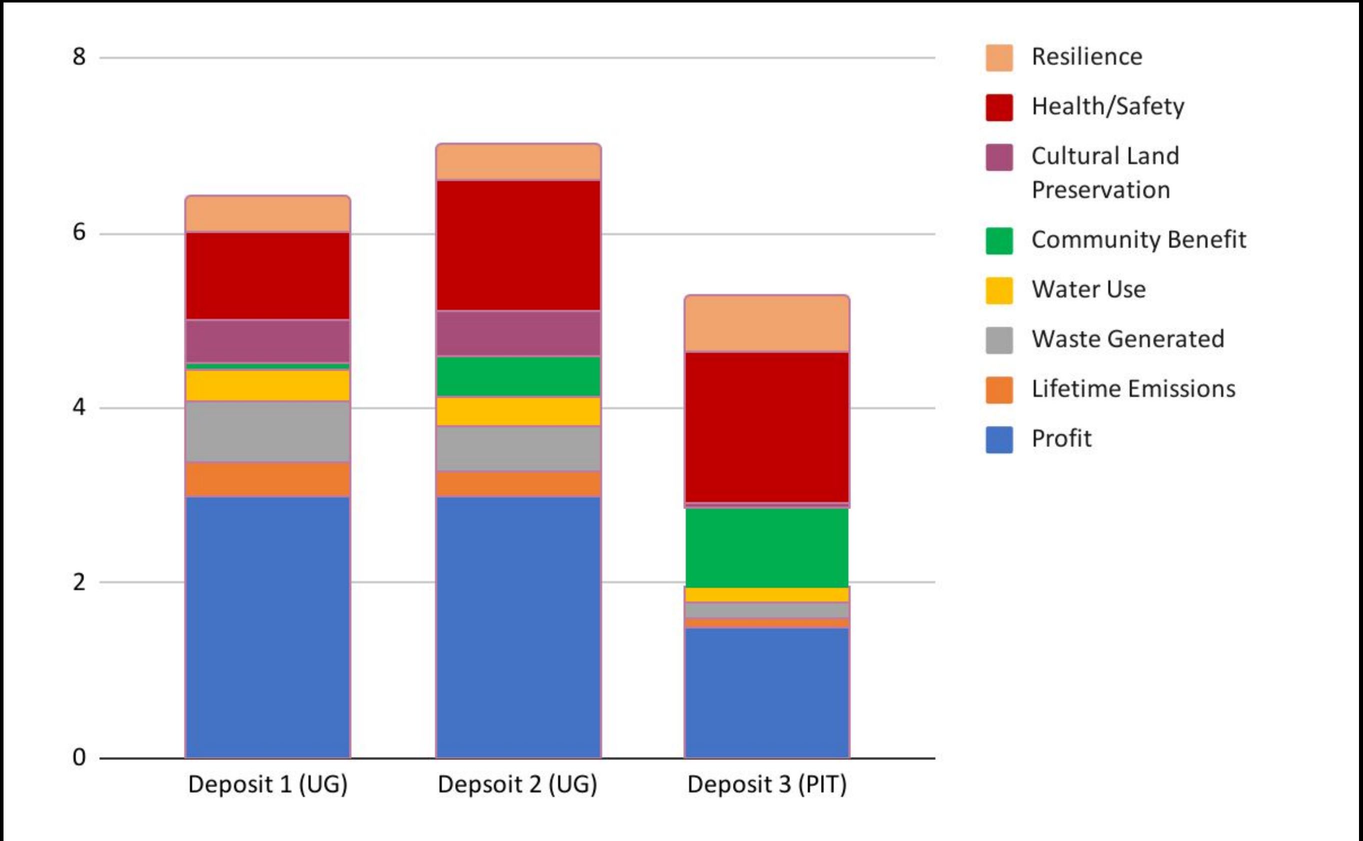
Deposit 3



Weighted Decision Matrix



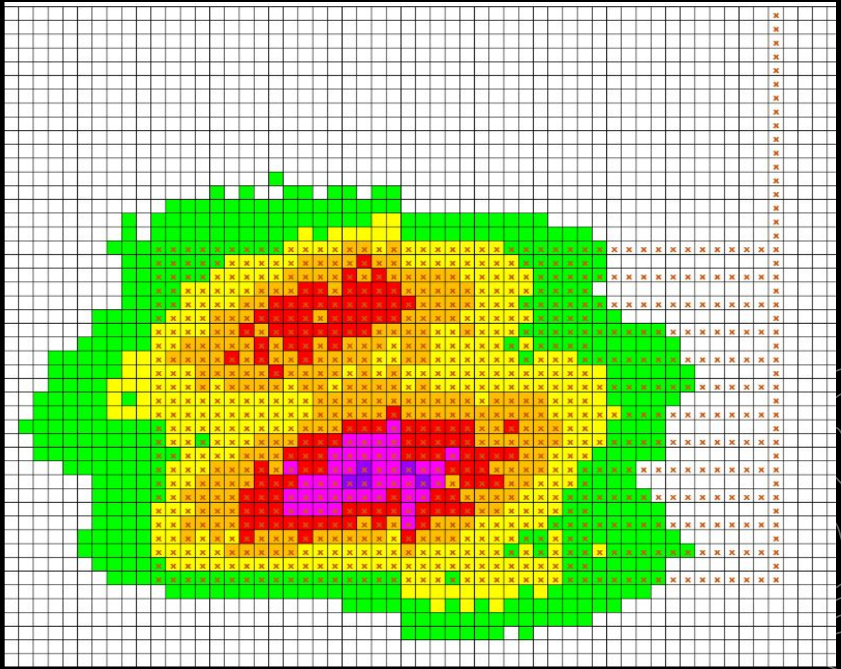
Comparisons/Defensible



Final Recommendation of Deposits

Final Recommendation: **Deposit 2 Underground Mine**

- **Economically:**
  - Over $8.2B in profit (community and Copper-Nicus Ltd)
  - Job creation and training
- **Environmentally:**
  - 8.5x less waste than open pit
  - **$970M Environmental Support Fund**
- **Socially:**
  - Minimal change to original way of life.
  - New community infrastructure



Why not Deposit 1 or Deposit 3?

Deposit 1

- Remote location
- Major new road, bridge, and power infrastructure required
- High development cost and uncertainty
- Fewer local benefits

Deposit 3

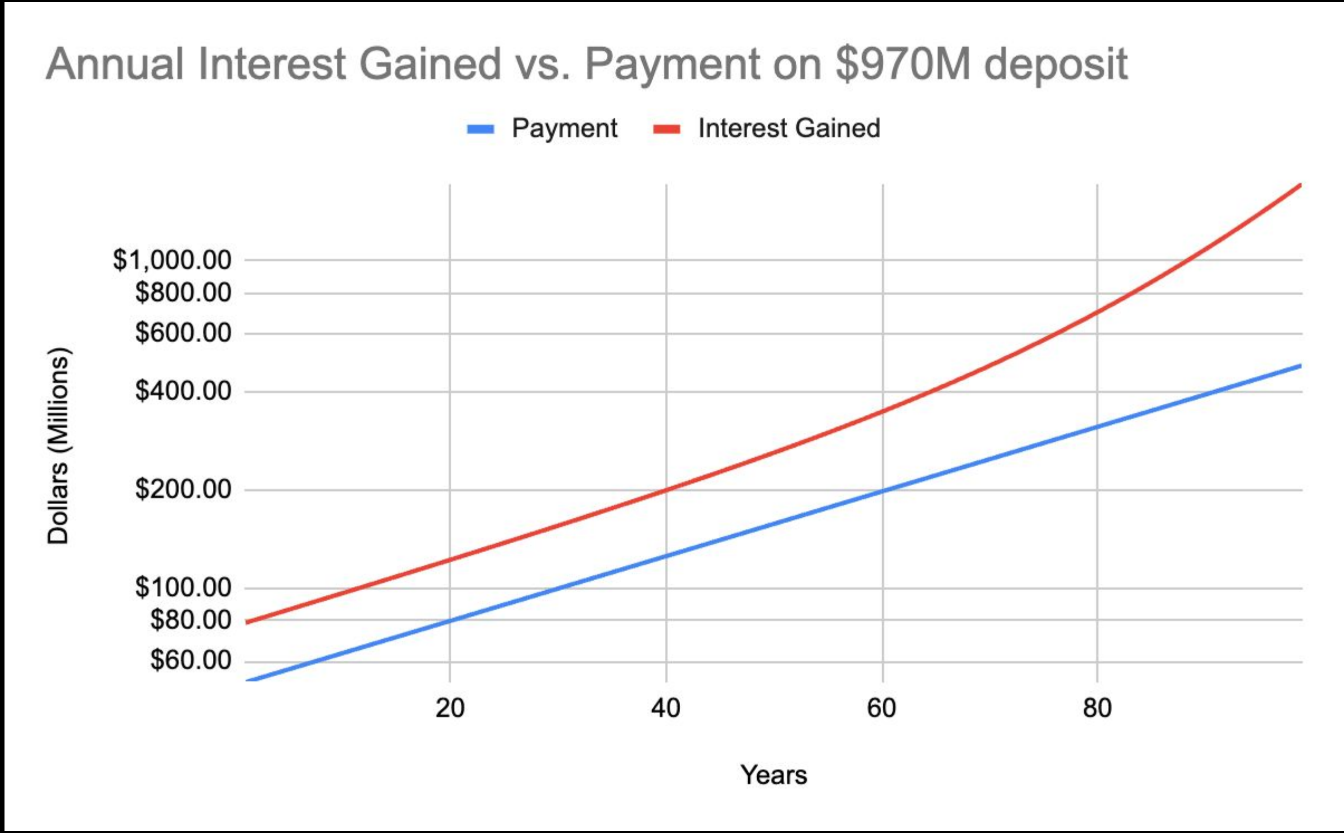
- Open pit mine with larger environmental footprint
- Close to archaeological sites and culturally significant land.
- Concerns about tailings storage and closure impacts

Thank You!

Generative AI

- GenAI tools were used to edit spelling and grammar of content.
- GenAI tools were used to make a ‘Copper-Nicus’ logo and stakeholder chart

Appendix: Interest on Economic Fund Deposit



Appendix: Raw WDM weighting and rationale

Criteria	Weighting	Deposit 1	Deposit 2	Deposit 3	WDM Assumptions
Profit from Mine (Initial from Block Model, supporting infrastructure needed to be accounted)	0.3	10	10	5	-Using the block model is how we found the profit for each mine, also taking into account the roads to be built and equipment needed to be bought. The further away, the more roads needed to be built subtracting from profit
Lifetime Emissions (SLCA Score)	0.05	8	6	2	-Uses scores from SLCA
Lifetime Waste generation (SLCA Score)	0.1	5	5	2	-Uses scores from SLCA
Water use (SLCA Score)	0.05	7	7	3	-Uses scores from SLCA
Benefit to the community (money, jobs, useful infractructure improvements)	0.1	0.8	4.7	9.2	Based on total cost of the mine divided by the distance from the community to it.
Preservation of cultrually significant land	0.05	10	10	1	Encompasses archeological sites/land traditional to indigenous peoples.
Lifetime Health and Safety (SLCA Score)	0.25	4	6	7	-Uses scores from SLCA
Ease of Resilience	0.1	4	4	6	Being closer to community is both an advantage and disadvantage, you can rally the community but you also affect them more. For the underground options, having a second channel makes the mines more resilient. May also include potential cost of building a second shaft on the other side of the deposit.

Appendix: Calculated WDM weighting and rationale

Caclulated Scores (this section will be automatically generated based on the values above)	Profit from Mine (Initial from Block Model, supporting infrastructure needed to be accounted)	Pofit	3	3	1.5
	Lifetime Emissions (SLCA Score)	Lifetime Emissions	0.4	0.3	0.1
	Lifetime Waste generation (SLCA Score)	Waste Generated	0.7	0.5	0.2
	Water use (SLCA Score)	Water Use	0.35	0.35	0.15
	Benefit to the community (money, jobs, useful infractructure improvements)	Community Benefit	0.08	0.47	0.92
	Preservation of cultrually significant land	tural Land Preservat	0.5	0.5	0.05
	Lifetime Health and Safety (SLCA Score)	Health/Safety	1	1.5	1.75
	Ease of Resilience	Resilience	0.4	0.4	0.6
			0	0	0
			0	0	0
			0	0	0
			0	0	0
			0	0	0
		Total	**6.43**	**7.02**	**5.27**